

Transformer-Empowered Invariant Grounding for Video Question Answering

Yicong Li¹, Xiang Wang¹, Junbin Xiao¹, Wei Ji¹, and Tat-Seng Chua¹

Abstract—Video Question Answering (VideoQA) is the task of answering questions about a video. At its core is the understanding of the alignments between video scenes and question semantics to yield the answer. In leading VideoQA models, the typical learning objective, empirical risk minimization (ERM), tends to over-exploit the spurious correlations between question-irrelevant scenes and answers, instead of inspecting the causal effect of question-critical scenes, which undermines the prediction with unreliable reasoning. In this work, we take a causal look at VideoQA and propose a modal-agnostic learning framework, named Invariant Grounding for VideoQA (IGV), to ground the question-critical scene, whose causal relations with answers are invariant across different interventions on the complement. With IGV, leading VideoQA models are forced to shield the answering from the negative influence of spurious correlations, which significantly improves their reasoning ability. To unleash the potential of this framework, we further provide a Transformer-Empowered Invariant Grounding for VideoQA (TIGV), a substantial instantiation of IGV framework that naturally integrates the idea of invariant grounding into a transformer-style backbone. Experiments on four benchmark datasets validate our design in terms of accuracy, visual explainability, and generalization ability over the leading baselines. Our code is available at <https://github.com/yl3800/TIGV>.

Index Terms—VideoQA, cross-modal visual reasoning, video-language, transformer, invariant learning.

I. INTRODUCTION

VIDEO Question Answering (VideoQA) aims to answer the natural language question regarding the video content. As a task of multi-modal reasoning, it is growing in popularity and importance to interactive AI, such as vision-language navigation for in-home robots and personal assistants [1], [2]. Clearly,

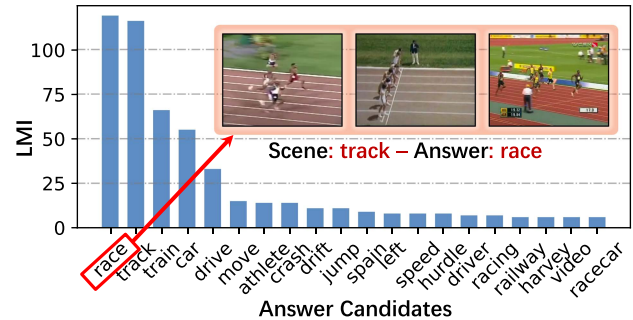
Manuscript received 14 December 2022; revised 1 July 2023; accepted 6 August 2023. Date of publication 9 August 2023; date of current version 3 October 2025. This research was supported in part by the NExT research, in part by the National Natural Science Foundation of China under Grant 9227010114, and in part by the University Synergy Innovation Program of Anhui Province under Grant GXXT-2022-040. Recommended for acceptance by Kristin Dana, Gang Hua, Stefan Roth, Dimitris Samaras, and Richa Singh. (Corresponding author: Xiang Wang.)

Yicong Li, Junbin Xiao, Wei Ji, and Tat-Seng Chua are with the National University of Singapore, Singapore 119077 (e-mail: liyicong@u.nus.edu; junbin@comp.nus.edu.sg; jiwei@nus.edu.sg; dscts@nus.edu.sg).

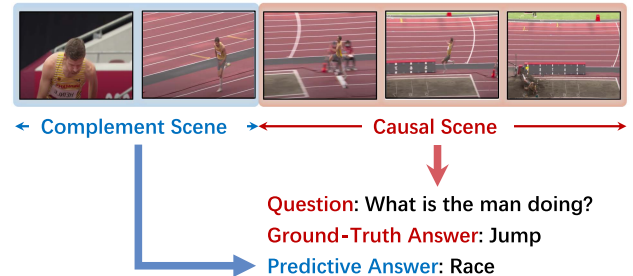
Xiang Wang is with the University of Science and Technology of China, Hefei 230026, China, and also with the Institute of Artificial Intelligence, Institute of Dataspaces, Hefei Comprehensive National Science Center, Hefei 230026, China (e-mail: xiangwang1223@gmail.com).

This article has supplementary downloadable material available at <https://doi.org/10.1109/TPAMI.2023.3303451>, provided by the authors.

Digital Object Identifier 10.1109/TPAMI.2023.3303451



(a) Local mutual information (LMI) between the “track” scene and answers.



(b) Running example of how the “track” complement deviates the answering.

Fig. 1. Running example. (a) Superficial correlations between visual scenes and answers; (b) Due to the spurious correlations, VideoQA model fails to answer the question.

inferring a reliable answer requires a deep understanding of visual scenes, linguistic semantics, and more importantly, the visual-linguistic alignments.

Recent VideoQA works [3], [4], [5], [6], [7], [8], [9] have shown continuous enthusiasm in architecture design, and the dominant paradigm can be epitomized as a combination of two modules: 1) a video-question encoder that encapsulates the visual content and linguistic semantics to multi-modal representations, and (2) answer decoder, which exploits these representations to model the visual-linguistic alignment and yield an answer. For model optimization, the criterion of empirical risk minimization (ERM) is unanimously adopted as the learning objective, that is, minimizing the loss between the predictive answer and the ground-truth answer.

However, the ERM criterion is prone to over-exploiting the superficial correlations between the video-question pairs and the answers. Specifically, we use the metric of local mutual information (LMI) [10] to quantify the correlations between

the “track” scene and answers. As Fig. 1(a) shows, most videos with “track” scene are associated with the “race” answer. Instead of inspecting the visual-linguistic alignments (i.e., which scene is critical to answer the question), ERM blindly captures all statistical relations. As Fig. 1(b) shows, it makes VideoQA model naively link the “track”-relevant videos with the strongly-correlated “race” answer, instead of the gold “jump” answer. Taking a causal look [11], [12] at VideoQA (see Section III), we partition the visual scenes into two parts: (1) *causal scene*, which holds the question-critical information, and (2) *its complement*, which is irrelevant to the answer. We scrutinize that the complement is spuriously correlated with the answer, thus ERM can hardly differentiate the effects of causal and complement scenes on the answer. Worse still, the unsatisfactory reasoning deviates the VideoQA models from some intriguing properties:

- *Visual-explainability* to exhibit “Which visual scene are the right reasons for the right answering?” [13], [14]. Taking Fig. 1(b) as an example to answer “What is the man doing?”, the model should attend the “jump” event presents in the last three clips, rather than referring to the “track” complement in the first two clips. One straightforward solution is “learning to attend” [15], [16], [17] to ground some scenes via the attentive mechanism. Nonetheless, guided by ERM, such attentive grounding still suffers from the spurious correlations, thus making the highly-correlated complement grounded.
- *Introspective learning* to double-check “How would the predictive answer change if the causal scenes were absent?”. On top of attentive grounding, the model needs to introspect whether the learned knowledge (i.e., attended scene) reliably and faithfully reflects the logic behind the answering. Briefly put, it should fail to answer the question if the causal scenes were removed.
- *Generalization ability* to enquire “How would the predictive answer response to the change of spurious correlations?”. As spurious correlations poorly generalize to open-world scenarios, the model should instead latch on the causal visual-linguistic relations that are stable across different environments.

Inspired by recent invariant learning [18], [19], [20], we conjecture that invariant grounding is the key to distinguishing causal scenes from the complements and overcoming these limitations. By “invariant”, we mean that the relations between question-critical scenes and answers are invariant regardless of changes in complements. Towards this end, we propose a new learning framework, *Invariant Grounding for VideoQA (IGV)*. Concretely, it integrates two additional modules into the VideoQA backbone model: a grounding indicator, a scene intervener. Specifically, the grounding indicator learns to attend the causal scenes for a given question and leaves the rest as the complement. Then, for the causal part of interest, the scene intervener conducts the causal interventions [11], [12] on its complement — that is, replace it with the stratification sampled from the other video instances and compose the “intervened videos”. After pairing the casual, complement, and intervened scenes with the question, we feed them into the backbone model to obtain the corresponding predictions: (1) causal prediction,

which approaches the gold answer, so as to achieve visual explainability; (2) complement prediction, which contains no critical clues to the ground-truth answer, thus enforces the backbone model to perform introspective reasoning; and (3) intervened prediction, which is consistent with the causal prediction across different intervened complements. Jointly learning these predictions enables the backbone model to alleviate the negative influence of multi-modal data bias.

Notably, IGV is a model-agnostic learning framework that improves the VideoQA backbones in a plug-and-play fashion [21]. Aside from the general framework studied in a preliminary version [21], this paper further extends IGV by presenting a *Transformer-Empowered instantiation (TIGV)*, which harmoniously integrates IGV into a Transformer [17] encoder-decoder architecture to perform a more effective invariant grounding.

Concretely, the *encoder* of TIGV performs cross-modal attention that absorbs the question semantics into every video scene. Meanwhile, the question sequence is prepended with a global question embedding (e.g., ‘[CLS]’) and a learnable token, whose semantic difference makes their interactions with the visual sequence a favorable indicator for the causal and complement scene, respectively. Therefore, this design naturally integrates the grounding indicator into the cross-modal encoder, whose output of casual, complement, and intervened representations will be directly fed to the decoder without pairing the question. For answer *decoding*, we draw inspiration from the recent triumph of query-based object detector [22], [23], and resort to a transformer-style answer decoder that utilizes the semantics of answer candidates to represent instances of interest, which are further leveraged as queries to decode the predictions from the casual, complement, and intervened scenes, respectively.

Apart from the instantiated TIGV architecture, we also explore superior model implementations. For example, during intervention, unlike our preliminary IGV framework [21] that relies on a global memory bank to sample the visual clips as intervened stratification, TIGV performs more efficient in-batch intervention on the encoded representation, i.e., the stratification for intervention contains the multi-modal information and is sampled from other video instances in the same mini-batch. This modification enables a more diverse sample space while saving memory without sacrificing accuracy. Besides, TIGV also exploits a more advanced language encoder that steadily improves the performance in all cases.

Finally, we additionally extend our experiments to the Causal-VidQA [24] dataset that challenges visual evidence and commonsense reasoning. Due to this evidence reasoning nature, Causal-VidQA serves as a desirable test-bed to examine model’s causal inferencing ability. Moreover, we comprehensively analyze our TIGV’s strengths and limitations by providing some heuristic observations on the grounded causal and complement scene.

Our contributions are summarized as follows:

- We highlight the importance of grounding causal scenes from the complements to visual-explainability, generalization, and introspective learning of VideoQA models.
- We propose an invariant learning scheme IGV, which incorporates invariant grounding into the VideoQA

models to mitigate the shortcut effect induced by question-irrelevant scene, thus enhancing the reasoning ability.

- We present a powerful instantiation TIGV by incorporating IGV framework into a transformer encoder-decoder framework, which achieves superior performance with preferable efficiency.
- On four benchmark datasets (i.e., NExT-QA [25], Causal-VidQA [24], MSRVTT-QA [5], MSVD-QA [5]), we conduct extensive experiments to justify the superiority of our method. The results demonstrate that TIGV significantly outperforms the previous state-of-the-arts.

II. PRELIMINARIES

In this section, we summarize the common paradigm of VideoQA models. Throughout the paper, we denote the random variables and their deterministic values by upper-cased (e.g. V) and lower-cased (e.g. v) letters, respectively.

Modeling: Given the video-question pair (V, Q) , the primer task of VideoQA is to generate an answer $\hat{A}$ as

$$\hat{A} = f_{\hat{A}}(V, Q), \quad (1)$$

where $f_{\hat{A}}$ is the VideoQA model, which is typically composed of two modules: video-question encoder f_E , and answer decoder f_D , i.e., $f_{\hat{A}} = f_E \circ f_D$. Specifically, the encoder includes two components: (1) a video model, which encodes visual scenes of the target video as a visual representation, such as motion-appearance memory design [3], [6], structural graph representation [9], [26], [27], [28], hierarchical architecture [7], [8], [29]; and (2) a language model, which encapsulates semantics of the text into a linguistic representation, such as global/local representation of textual content [30], [31] and graph representation of grammatical dependencies [27]. On top of these representations, the decoder learns the visual-linguistic alignments to generate the answer. In particular, the alignments are modeled via cross-modal interaction like graph alignment [27], cross-attention [4], [26], [30] and co-memory [6], *etc.*

Learning: To optimize these modules, leading VideoQA models [3], [6], [26], [29], [30] cast the multi-modal reasoning problem as a supervised learning task and adopt the learning objective of empirical risk minimization (ERM) as

$$\min \mathcal{L}_{\text{ERM}}(\hat{A}, A), \quad (2)$$

where $\mathcal{L}_{\text{ERM}}$ is the risk function to measure the loss between the predictive answer $\hat{A}$ and the ground-truth answer A , which is usually set as cross-entropy loss [6], [29] or hinge loss [3], [26]. In essence, ERM encourages these VideoQA modules to capture the statistical correlations between the video-question pairs and the answers.

III. A CAUSAL LOOK AT VIDEOQA

From the perspective of causal theory [11], [12], we revisit the VideoQA scenario to show superficial correlations between video-question pairs and answers. We then analyze ERM's suffering from the spurious correlations.

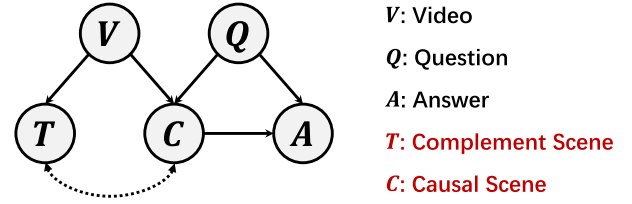


Fig. 2. Causal graph of VideoQA.

A. Causal Graph of VideoQA

In general, multiple visual scenes are present in a video. However, only part of the scenes are critical to answering the question of interest, while the rest hardly offers information relevant to the question. Moreover, the linguistic variations in different questions should activate different scenes of a video. These facts inspire us to split the video into the causal and complement parts in terms of the question. Here we use a causal graph [11], [12] to exhibit the relationships among five variables: input video V , input question Q , causal scene C , complement scene T , and ground-truth answer A . Fig. 2 illustrates the causal graph, where each link is a cause-and-effect relationship between two variables:

- $C \leftarrow V \rightarrow T$. The input video V consists of C and T . For example, the video in Fig. 1(b) is the combination of the first two clips (i.e., C) and the last three clips (i.e., T).
- $V \rightarrow C \leftarrow Q$. The causal scene C is conditional upon the video-question pair (V, Q) , which distills Q -relevant information from V . For a given V , the variations in Q result in different C .
- $Q \rightarrow A \leftarrow C$. The answer A is determined by the question Q and causal scene C , reflecting the visual-linguistic alignments. Considering the example in Fig. 1(b) again, C is the oracle scene that perfectly explains why “jump” is labeled as the ground truth to answer the question.
- $T \dashleftarrow C$. The dashed arrow summarizes the additional probabilistic dependencies [32] between C and T . Such dependencies are usually caused by the selection bias or inductive bias during the process of data collection or annotation [33], [34]. For example, one mostly collects the videos with the “jump” events on the “track”. Here we list three typical scenarios: (1) C is independent of T (i.e., $T \perp C$); (2) C is the direct cause of T (i.e., $C \rightarrow T$), or vice versa (i.e., $C \leftarrow T$); (3) C and T have a common cause E (i.e., $C \leftarrow E \rightarrow T$). See Appendix Section B for details, available online.

B. Spurious Correlations

Taking a closer look at the causal graph, we find that the complement scene T and the ground-truth answer A can be spuriously correlated. Specifically, as the confounder [11], [12], [32] between T and A , Q and V open the backdoor paths: $T \leftarrow V \rightarrow C \rightarrow A$ and $T \leftarrow V \rightarrow C \leftarrow Q \rightarrow A$, which make T and A spuriously correlated even though there is no direct causal path from T to A . Worse still, $T \dashleftarrow C$ can amplify this issue. Assuming $C \rightarrow T$, C becomes an additional confounder to yield

another backdoor path $T \leftarrow C \rightarrow A$. Such spurious correlations can be summarized as the probabilistic dependence: $A \not\perp T$.

As ERM naively captures the statistical correlations between video-question pairs and answers, it fails to distinguish the causal scene C and complement scene T , thus failing to mitigate the negative influence of spurious correlations. As a result, it limits the reasoning ability of VideoQA models, especially in the following aspects: (1) visual-explainability to reason about “Which visual scenes are the supporting evidence to answer the question?”; (2) introspective learning to answer “How would the answer change if the causal scenes were absent?”; and (3) generalization ability to enquire “How would the answer response to the change of spurious correlations?”.

IV. METHODOLOGY

We get the inspiration from invariant learning [18], [19], [20] and argue that invariant grounding of causal scenes is the key to reducing the spurious correlations. Specifically, we first present our learning framework, Invariant Grounding for VideoQA (*IGV*) to overcome the foregoing limitations of the existing VideoQA models. We then present TIGV, a transformer-empowered IGV instantiation, to unleash the potential of the invariant grounding.

A. Invariant Grounding for VideoQA

Upon closer inspection on the causal graph, we notice that the ground-truth answer A is independent of the visual complement T , only when conditioned on the question Q and the causal scene C , more formally

$$A \perp T \mid C, Q. \quad (3)$$

This probabilistic independence indicates the invariance — that is, the relations between the (C, Q) pair and A are invariant regardless of changes in T . The causal relationship $Q \rightarrow A \leftarrow C$ is invariant across different T . Taking Fig. 1(b) as an example, if the question and the causal scene (i.e., the last three clips) remain unchanged, the answer should arrive at “jump”, no matter how the complement scene varies¹ (e.g. substitute the “track” clips by the “cloud”- or “sea”-relevant ones). This highlights that the (C, Q) pair is the key to shielding A from the influence of T .

Modeling: However, only the (V, Q) pair and A are available in the training set, while neither C nor the grounding function towards C is known. This motivates us to incorporate visual grounding into the VideoQA modeling, where the grounded scene $\hat{C}$ aims to estimate the oracle C and guide the prediction of answer $\hat{A}$. More formally, instead of the conventional modeling (1), we systematize the modeling process as

$$\hat{C} = f_{\hat{C}}(V, Q), \quad \hat{A} = f_{\hat{A}}(\hat{C}, Q), \quad (4)$$

where $f_{\hat{C}}$ is the grounding model, and $f_{\hat{A}}$ is the VideoQA model that relies on the $(\hat{C}, Q)$ pair instead. See Sections IV-B and IV-C for our implementations of $f_{\hat{C}}$ and $f_{\hat{A}}$.

¹Note that the complement substitutes will not involve the question-relevant scenes, in order to avoid creating additional paths from T to A .

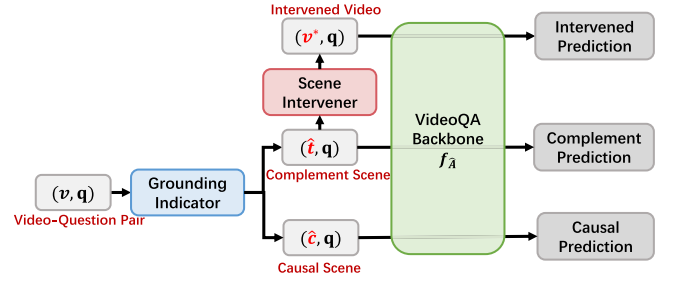


Fig. 3. Overview of IGV framework with the off-the-shelf VideoQA Model.

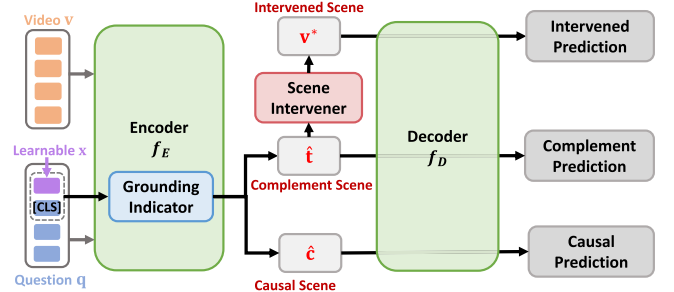


Fig. 4. Overview of TIGV.

Learning: Nonetheless, simply integrating visual grounding with the VideoQA model falls into the “learning to attend” paradigm, which still suffers from the spurious correlations and erroneously attends to the complement scenes as $\hat{C}$. To this end, we exploit the invariance property of C (3) and reformulate the learning objective of invariant grounding as

$$\min_{f_{\hat{A}}, f_{\hat{C}}} \mathcal{L}_{IGV}(\hat{A}, A), \quad \text{s.t. } A \perp \hat{T} \mid \hat{C}, Q, \quad (5)$$

where $\mathcal{L}_{IGV}$ is the loss function to our IGV; $\hat{T} = V \setminus \hat{C}$ is the complement of $\hat{C}$.

Overview: An overview of the IGV framework [21] is illustrated in Fig. 3. It involves two additional modules beyond the VideoQA backbone model, a grounding indicator and a scene intervener. Concretely, the grounding indicator aims to identify the causal and complement parts from the video scene ($\hat{c}$ and $\hat{t}$), while the scene intervener takes as input the causal scene and yields the intervened counterpart v^* . Then, the three obtained scenes are paired with question q then fed into the backbone model $f_{\hat{A}}$. As a result, the output predictions are carefully designed to approach different targets, which enables the proposed invariant grounding. In the next section, we introduce how to incorporate IGV into a transformer encoder-decoder architecture, thus achieving the transformer-empowered instantiation, TIGV.

B. Transformer-Empowered IGV (TIGV)

As shown in Fig. 4, TIGV instantiates the invariant grounding framework by absorbing the grounding indicator into the encoder. Moreover, it performs scene intervention on the encoded visual scene, where the resulting scenes can be directly fed to a transformer decoder, thus achieving a more efficient invariant grounding.

1) *Encoder and Grounding Indicator*: Given a paired video-question instance (v, q) , we first split v as a sequence of K fixed clips, and then feed it to a off-the-shelf video model M_{vid} to obtain visual embeddings. For the question q of L words, we prepend it with a special [CLS] token and feed the sequence to a off-the-shelf language model M_{lan} to generate the linguistic embeddings, where the prepended [CLS] token is able to capture the global semantic of the question [35]. Formally, we acquire the encoded video tokens $\mathbf{v} \in \mathbb{R}^{K \times d}$ and textual tokens $[\mathbf{q}_{cls}; \mathbf{q}] \in \mathbb{R}^{(1+L) \times d}$ as

$$\mathbf{v} = \text{MLP}_{vid}(M_{vid}(v)), \quad (6)$$

$$[\mathbf{q}_{cls}; \mathbf{q}] = \text{MLP}_{lan}(M_{lan}([\text{CLS}; q])), \quad (7)$$

where MLP_{vid} and MLP_{lan} are multilayer perceptions that project video and question embeddings into a common d -dimensional space, respectively, And $[\cdot; \cdot]$ refers to the concatenation operation.

To naturally vest the encoder with the ability of scene grounding, we first prepend a learnable embedding $\mathbf{x}$ to the textual tokens, and feed it to the encoder together with video tokens to realize a visual-language cross-attention, where video tokens are treated as the query, and the extended question tokens are used as key and value. As a result, this yields encoded visual token $\mathbf{v}' \in \mathbb{R}^{K \times d}$ and pre-normalized attention score $\mathbf{z} \in \mathbb{R}^{K \times (2+L)}$ as

$$\mathbf{v}_{out}, \mathbf{z} = \text{MHA}(\mathbf{v}, [\mathbf{x}; \mathbf{q}_{cls}; \mathbf{q}]), \quad (8)$$

$$\mathbf{v}' = \text{MLP}(\text{LN}(\mathbf{v}_{out})) + \mathbf{v}_{out}, \quad (9)$$

where MHA, LN denotes Multi-Head Attention and Layer Normalization, respectively. Notably, the prepended question tokens $[\mathbf{x}; \mathbf{q}_{cls}]$ are only used to compute attention score $\mathbf{z}$, and their effect is masked out when calculating $\mathbf{v}'$.

To indicate the causal and complement scene from $\mathbf{v}'$, we exploit the first two columns of $\mathbf{z}$ (i.e., $[\mathbf{z}_x; \mathbf{z}_{cls}]$) that denotes interactions between the prepended token (i.e., $[\mathbf{x}; \mathbf{q}_{cls}]$) and all video tokens. Then their intensities are normalized as $\mathbf{p}_{\hat{c}} \in \mathbb{R}^K$ and $\mathbf{p}_{\hat{t}} \in \mathbb{R}^K$ via a softmax to indicate the importance of each visual clip

$$[\mathbf{p}_{\hat{c}}; \mathbf{p}_{\hat{t}}] = \text{Softmax}([\mathbf{z}_{cls}; \mathbf{z}_x]). \quad (10)$$

Intuitively, $\mathbf{q}_{cls}$ contains the global question semantics, which should exhibit higher interaction intensity with the causal scene. Thus, we use $\mathbf{p}_{\hat{c}} \in \mathbb{R}^K$ to denote the probability of each clip belonging to the causal scene $\hat{c}$, while $\mathbf{p}_{\hat{t}} \in \mathbb{R}^K$ is in contrast to $\mathbf{p}_{\hat{c}}$ to show how likely each clip composes the complement $\hat{t}$. However, such soft masks generated via attentive mechanism makes $\hat{c}$ and $\hat{t}$ overlapped, which is unable to shield the answering from the influence of the complement. Hence, the grounding indicator produces discrete selections instead to make $\hat{c}$ and $\hat{t}$ disjoint. Nonetheless, simple sampling or selection is not differentiable. To achieve differentiable discrete selection, we apply Gumbel-Softmax [36]

$$\mathbf{I} = \text{Gumbel-Softmax}([\mathbf{p}_{\hat{c}}; \mathbf{p}_{\hat{t}}], \tau), \quad (11)$$

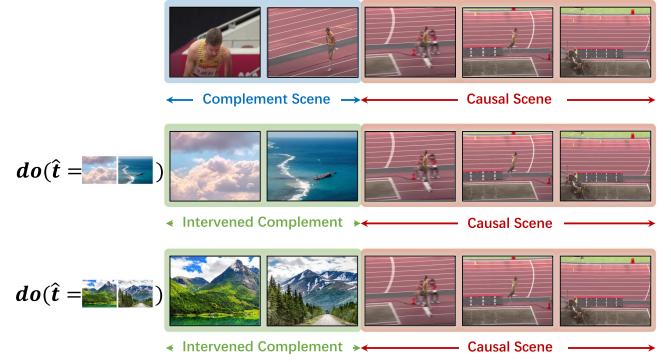


Fig. 5. Illustration of interventional distribution.

where the first and second column of the output indicator vector $\mathbf{I} \in \mathbb{R}^{K \times 2}$ index $\hat{t}$ and $\hat{c}$ over K encoded video tokens, respectively. The temperature τ is a hyper-parameter. As such, we can devise $\hat{t}$ and $\hat{c}$ as follows:

$$\hat{t} = \{I_{k0} \cdot \mathbf{v}'_k | I_{k0} = 1\}, \quad \hat{c} = \{I_{k1} \cdot \mathbf{v}'_k | I_{k1} = 1\}, \quad (12)$$

where I_{k0} and I_{k1} suggest that the k th encoded video token belongs to the causal and complement scenes, respectively.

2) *Scene Intervener*: It is challenging to learn the grounding indicator, owing to the lack of supervisory signals of clip-level importance. To remedy this issue, we propose the scene intervener, which preserves the estimated causal scene $\hat{c}$ but intervenes the estimated complement $\hat{t}$ to create the “intervened scene”, as Fig. 5 shows.

Specifically, for the a sequence of encoded video tokens $\mathbf{v}' = \hat{c} \cup \hat{t}$, the intervener conducts causal interventions [11], [12] on its $\hat{t}$ — that is, random sample complement tokens $\hat{t}^*$ from other training instances in the same mini-batch to replace $\hat{t}$ and combine it with $\hat{c}$ at hand as a new sequence $\mathbf{v}^* = \hat{c} \cup \hat{t}^*$.

It is worthwhile mentioning that, compared with our preliminary intervention strategy in IGV that needs to maintain a massive memory bank on visual clips, TIGV performs a much more efficient in-batch, where the stratification for intervention is sampled from other instances in the same mini-batch instead of the whole training set. Moreover, since the TIGV conducts intervention on encoded visual tokens, the sampling space is naturally expanded by embracing the combination of semantic information and visual clips, which creates more diverse intervened scenes. Notably, distinct from current invariant learning studies [18], [19], [20] that only partition the training set into different environments, our scene intervener exploits the interventional distributions [37] on encoded video instead. The interventional distribution (i.e., , the token sequence with the same interventions) can be viewed as one environment.

3) *Decoder*: To better exploit the answer information, we discard conventional designs that formulate the answer decoder f_D as either a linear alignment [7], [8], [29] or a cross-modal matching [9]. Instead, we resort to a query-based transformer to align the grounded scene to the answer space, where an answer query is formed to carry the answer information. According to the QA setting, the answer candidates are either given as $|A_{mc}|$

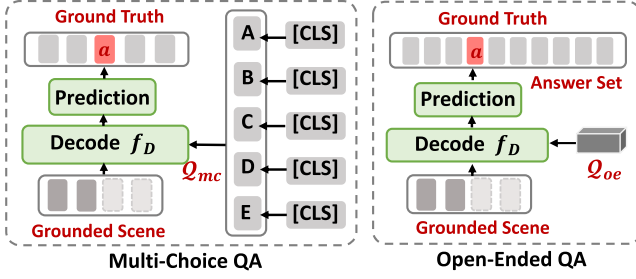


Fig. 6. Illustration of TIGV's decoder.

sentences that are tailored for each instance, (i.e., Multi-Choice QA), or a global answer set of $|A_{oe}|$ categories that are shared among all instances, (i.e., Open-Ended QA). As shown in Fig. 6, the answer query has two formations according to the task settings.

Multi-Choice QA: In this setting, we first prepend a [CLS] token to each of $|A_{mc}|$ answer candidates, then using the same language model M_{lan} in the question encoding (cf. (7)) to acquire $|A_{mc}|$ holistic representations. Next, we concatenate them to form the answer query $Q_{mc} \in \mathbb{R}^{|A_{mc}| \times d}$. During decoding, we feed the transformer decoder with Q_{mc} as the target sequence and grounded scene $\mathbf{h} \in \{\hat{\mathbf{c}}, \hat{\mathbf{t}}, \mathbf{v}^*\}$ as memory, which yields the decoded sequence $\mathbf{h}_{mc} \in \mathbb{R}^{|A_{mc}| \times d}$ as

$$\mathbf{h}_a = \text{MLP}(\text{LN}(\text{MHSA}(Q_{mc}))) + \text{MHSA}(Q_{mc}), \quad (13)$$

$$\mathbf{h}_{mc} = \text{MLP}(\text{LN}(\text{MHA}(\mathbf{h}_a, \mathbf{h}))) + \text{MHA}(\mathbf{h}_a, \mathbf{h}), \quad (14)$$

where MHSA denotes the Multi-Head Self-Attention operation. It is worth noticing that, since the correctness of an answer candidate is irrelevant to its position, the answer query is fed to the decoder without position encoding. Finally, a linear projection is applied to the decoded sequence $\mathbf{h}_{mc}$ to produce the prediction $\hat{a}_{mc} \in \mathbb{R}^{|A_{mc}|}$.

$$\hat{a}_{mc} = \text{MLP}(\mathbf{h}_{mc}). \quad (15)$$

Open-Ended QA: By convention [8], [29], each answer candidate in open-ended QA is treated as a category without semantic meaning, which composes a global answer set that is much bigger than multi-choice QA (i.e., $|A_{oe}| \gg |A_{mc}|$). Therefore, we initialize the answer query as a single learnable embedding $Q_{oe} \in \mathbb{R}^d$ to acquire the answer information in decoding. Then, Analogous to (13)–(14), we feed Q_{oe} to the decoder f_D together with the grounded scene to attain the decoded representation h_{oe} . Finally, the prediction $\hat{a}_{oe} \in \mathbb{R}^{|A_{oe}|}$ is acquired by projecting h_{oe} to the answer space

$$\hat{a}_{oe} = \text{MLP}(h_{oe}). \quad (16)$$

4) **Joint Training:** For a video-question pair instance (v, q) , we have embedded the question semantic to the visual token $\mathbf{v}$ and established its corresponding causal scene $\hat{\mathbf{c}}$, complement scene $\hat{\mathbf{t}}$, and intervened scene $\mathbf{v}^*$ via the grounding indicator and scene intervener. We next feed these instances into the decoder f_D to obtain three predictions:

- **Causal prediction:** As the causal scene $\hat{\mathbf{c}}$ is expected to be sufficient and necessary to answer the question,

we leverage its predictive answer $f_D(\hat{\mathbf{c}})$ to approach the ground-truth answer a solely

$$\mathcal{L}_{\hat{\mathbf{c}}} = \text{XE}(f_D(\hat{\mathbf{c}}), a), \quad (17)$$

where XE denotes the cross-entropy loss.

- **Complement prediction:** As no critical clues should exist in the complement scene $\hat{\mathbf{t}}$ to answer the question q , we encourage its predictive answer $f_D(\hat{\mathbf{t}})$ to evenly predict all answers. This uniform loss is formulated as

$$\mathcal{L}_{\hat{\mathbf{t}}} = \text{KL}(f_D(\hat{\mathbf{t}}), u), \quad (18)$$

where KL denotes KL-divergence, and u is the uniform distribution over all answer candidates.

- **Intervened prediction:** According to the invariant constraint (3), the causal relationship between the causal scene and the answer is stable across different complements. To parameterize this constraint, we enforce the intervened scene $\mathbf{v}^*$ to hold the consistent predictions with the causal scene $\hat{\mathbf{c}}$

$$\mathcal{L}_{\mathbf{v}^*} = \mathbb{E}_{\hat{\mathbf{t}}^* \in \mathcal{T}} (\text{KL}(f_D(\mathbf{v}^*), f_D(\hat{\mathbf{c}}))). \quad (19)$$

Aggregating the foregoing risks, we attain the learning objective of IGV

$$\mathcal{L}_{\text{IGV}} = \mathbb{E}_{(v, q, a) \in \mathcal{O}^+} \mathcal{L}_{\hat{\mathbf{c}}} + \mathcal{L}_{\hat{\mathbf{t}}} + \mathcal{L}_{\mathbf{v}^*}, \quad (20)$$

where $\mathcal{O}^+$ is the training set of the video-question pair (v, q) and the ground-truth answer a ; Jointly learning these predictions enables the TIGV to uncover the question-critical scene, so as to mitigate the negative influence of spurious correlations between the question-irrelevant complement scene and answer. In the inference phase, we use the causal prediction $f_D(\hat{\mathbf{c}})$ to answer the question.

C. Difference Between IGV and TIGV

Recall that IGV is a model-agnostic learning framework that can be applied to any off-the-shelf VideoQA models without bells and whistles. In this section, we briefly recap the preliminary implementation of IGV [21] (instantiated by off-the-shelf VideoQA models), by emphasizing its difference with the aforementioned TIGV instantiation.

Video-Question Encoding: For a given video-question instance (v, q) , we encode v and q using a similar video model M_{vid} and language models M_{lan} as (6) and (7), which yield the video and question embedding as a sequence of K fixed visual clips and L language tokens, respectively. Then two independent LSTMs [38] are employed as follow:

$$\mathbf{v}_g, \mathbf{v}_l = \text{LSTM}_v(M_{vid}(v)), \quad (21)$$

$$\mathbf{q}_g, \mathbf{q}_l = \text{LSTM}_q(M_{lan}(q)), \quad (22)$$

LSTM_v outputs $\mathbf{v}_l \in \mathbb{R}^{K \times d}$ as the local representations of clips, and yields the last hidden state $\mathbf{v}_g \in \mathbb{R}^d$ as the global representations of the holistic video. Analogously, LSTM_q generates $\mathbf{q}_l \in \mathbb{R}^{L \times d}$ as the local representations of tokens, and makes the last hidden state $\mathbf{q}_g \in \mathbb{R}^d$ represent the question holistically, here d is the hidden dimension. Distinct from TIGV that leverages a [CLS] token to acquire global question embedding, IGV

demands an additional LSTM to encapsulate the question from a holistic view.

Grounding Indicator: Compare with the integrated grounding indicator in TIGV, IGV [21] implements the isolated grounding indicator via two additional attention scores $\mathbf{p}_{\hat{c}}, \mathbf{p}_{\hat{t}} \in \mathbb{R}^K$ that indicates the probability of each visual clip being a causal or a complement scene, respectively

$$\mathbf{p}_{\hat{c}} = \text{Softmax}(\text{MLP}_1(\mathbf{v}_l) \cdot \text{MLP}_2(\mathbf{q}_g)^\top), \quad (23)$$

$$\mathbf{p}_{\hat{t}} = \text{Softmax}(\text{MLP}_3(\mathbf{v}_l) \cdot \text{MLP}_4(\mathbf{q}_g)^\top). \quad (24)$$

Similar to (11) and (12), we apply a Gumbel-Softmax upon the concatenation of $\mathbf{p}_{\hat{c}}$ and $\mathbf{p}_{\hat{t}}$ (i.e., $[\mathbf{p}_{\hat{c}}, \mathbf{p}_{\hat{t}}] \in \mathbb{R}^{K \times 2}$), where the first and second column of the outputs indicator vector $\mathbf{I} \in \mathbb{R}^{K \times 2}$ are used to indexes $\hat{c}$ and $\hat{t}$ over k clips, respectively. It is worth noticing that IGV [21] uses the same query $\mathbf{q}_g$ with different transformations to identify the casual and complement scenes, while TIGV uses different queries (i.e., $\mathbf{q}_{cls}$ and $\mathbf{x}$ in (8)) to explicitly model the semantic difference between causal and complement scenes. Furthermore, the invariant grounding in IGV is performed on the pure (uni-modal) video-clip representations. In contrast, TIGV integrates the grounding indicator into the cross-modal encoding, and thus performs invariant grounding on the encoded visual tokens that are already enclosed with question semantics.

Scene Intervener: Different from TIGV that carries out in-batch intervention on the encoded visual tokens, IGV first collects visual clips from other training videos as a memory bank of complement stratification, $\hat{\mathcal{T}} = \{\hat{t}\}$. Then, it intervenes the target video $v = \hat{c} \cup \hat{t}$ by replacing $\hat{t}$ with a random sample a complement stratification $\hat{t}^* \in \hat{\mathcal{T}}$. As a result, IGV acquires the intervened scene as a new video $v^* = \hat{c} \cup \hat{t}^*$, instead of assembled visual tokens in TIGV.

Joint Learning: As a comparison, the grounded scenes in TIGV are presented as the encoded visual token that already entails question semantics, thus, it only needs a decoder to align the grounded scenes to predictions. However, to facilitate backbone-agnostic modeling, IGV needs to first pair the question q with the causal scene $\hat{c}$, complement scene $\hat{t}$, and intervened video v^* to synthesize three new instances, $(\hat{c}, q)$, $(\hat{t}, q)$, (v^*, q) , and then feed these instances into the entire backbone VideoQA model $f_{\hat{A}}$ to obtain the corresponding three predictions.

$$\mathcal{L}_{\hat{c}} = \text{XE}(f_{\hat{A}}(\hat{c}, q), a), \quad (25)$$

$$\mathcal{L}_{\hat{t}} = \text{KL}(f_{\hat{A}}(\hat{t}, q), u), \quad (26)$$

$$\mathcal{L}_{v^*} = \mathbb{E}_{\hat{t}^* \in \hat{\mathcal{T}}}(\text{KL}(f_{\hat{A}}(v^*, q), f_{\hat{A}}(\hat{c}, q))). \quad (27)$$

Analogous to (20), the final IGV learning objective is obtained by summing the foregoing risks.

V. EXPERIMENTS

Datasets: To justify the effectiveness of our design, we conducted extensive experiments on four datasets that challenge the various aspects of video understanding: For the prevailing Multi-Choice QA, *NExT-QA* [25] emphasizes the causal and temporal

TABLE I
DATASET STATISTICS. MC: MULTI-CHOICE QA. OE: OPEN-ENDED QA

Dataset	Challenge	#QA pair	V-Len	Q-Len	QA
NExT-QA	Causal & Temporal	48K	44s	11.6	MC
Causal-VidQA	Evidence & Commonsense	161K	9s	9.5	MC
MSVD-QA	Description	50K	10s	6.6	OE
MSRVTT-QA	Description	244K	15s	7.4	OE

interactions among visual objects, while *Causal-VidQA* [24] focuses on the evidence and commonsense reasoning. In terms of the Open-Ended QA, we adopt the widely-used *MSVD-QA* [5] and *MSRVTT-QA* [5] that highlights the recognition of the video elements, such as their attributes, actions as well as activities. For all four datasets, we follow the official data splits for experiments and report accuracy as evaluation metric. The related statistics are presented in Table III.

Implementation Details: For the visual feature, we follow previous works [8], [9], [26] and extract video instance as a sequence of $K=16$ clips, where each clip is encoded by the video model M_{vid} that consists of a pre-trained 3D ResNeXt-101 and ResNet-101, and the yielded motion and appearance feature are concatenated to represent a clip embedding. As for the textual embedding, TIGV adopts M_{lan} as a pretrained Deberta model [35], which is jointly finetuned with VideoQA backbone to encode the question and answer, while IGV implements M_{lan} as finetuned Bert model [39]. During training, we adopt the Adam optimizer with the initial learning rate of $1e-4$, which will be multiplied by 0.1 if no validation improvements in 2 epochs. We set the batch size to 256 and the maximum epoch to 30.

A. Comparisons With SoTA Methods

1) *Results on Causal-VidQA:* As shown in Table II, the performance of TIGV exceeds current SoTAs by a large margin, with questions of all sub-types surpassing their competitors, which demonstrates the strong generalization ability of invariant grounding. Specifically, we have two major observations:

First, we notice that TIGV brings more favorable gain to the explanatory question (Acc@E: +9.2) and predictive question (Acc@C: +6.7~11.9) than it does to the descriptive question (Acc@D: +3.9). The underlying explanations are as follows: 1) The explanatory question requires an explicit identification of the visual clue that is responsible for answering, which naturally fits the inherent causal reasoning ability of TIGV. By invariant grounding, TIGV is able to pinpoint the answer-critical video scene that encloses the required visual clue, thus bringing a much preferable improvement over leading baselines. 2) The counterfactual question generally challenges the commonsense reasoning ability of the VideoQA model. We attribute such strong performance gain to our exploitation of advanced language models, which naturally incorporate more commonsense knowledge into answering. 3) However, the descriptive questions simply inquire about the static attribute of the visual object, where a glimpse of several random frames might suffice the answering. Therefore, existing SoTAs can yield a reasonable

TABLE II
ACCURACY (%) COMPARISON ON CAUSAL-VIDQA. D: DESCRIPTION, E: EXPLANATION, P: PREDICTION, C: COUNTERFACTUAL. IF NOT OTHERWISE INDICATED, IGV USES HGA* AS THE DEFAULT BACKBONE THROUGHOUT THE EXPERIMENTS. SEE SECTION C FOR THE DETAILED ARCHITECTURE OF HGA*, AVAILABLE ONLINE

Methods	Causal-VidQA								Acc@All
	Acc@D	Acc@E	Acc@P			Acc@C			
			Q → A	Q → R	Q → AR	Q → A	Q → R	Q → AR	
HCRN [29]	56.4	61.6	51.7	51.3	32.6	51.6	53.4	32.7	48.1
HGA [26]	65.7	63.5	49.4	50.6	32.2	52.4	55.9	34.3	48.9
B2A [27]	66.2	62.9	49.0	50.2	31.2	53.3	56.3	35.2	49.1
IGV	63.8	71.4	45.9	47.8	30.6	63.4	62.8	45.8	52.9
TIGV	70.1	72.7	57.1	58.2	41.3	65.2	63.0	47.0	57.8
Imp. vs. SoTA	+3.9	+9.2	+5.4	+6.9	+8.7	+11.9	+6.7	+11.8	+8.7

TABLE III
ACCURACY (%) COMPARISON ON NExT-QA, MSVD-QA, AND MSRVTT-QA. ACC@C, T, D, DENOTE QUESTIONS TYPE OF CAUSAL, TEMPORAL, AND DESCRIPTIVE IN NExT-QA, RESPECTIVELY

Methods	NExT-QA				MSVD	MSRVTT
	Acc@C	Acc@T	Acc@D	Acc@All		
Co-Mem† [6]	45.9	50.0	54.4	48.5	34.6	35.3
HCRN [29]	47.1	49.3	54.0	48.9	36.1	35.6
HGA [26]	48.1	49.1	57.8	50.0	34.7	35.5
HOSTR [7]	-	-	-	-	39.4	35.9
PGAT [40]	-	-	-	-	39.0	38.1
MHN [41]	-	-	-	-	40.4	38.6
HQGA [8]	49.0	52.3	59.4	51.8	41.2	38.6
EIGV [42]	51.3	<u>53.1</u>	62.7	<u>53.7</u>	<u>40.8</u>	38.5
VGT [9]	<u>51.6</u>	51.9	63.7	<u>53.7</u>	-	<u>39.7</u>
IGV	48.6	51.7	59.6	51.3	40.8	38.3
TIGV	55.0	56.3	<u>62.9</u>	56.7	43.1	41.1
Imp. vs. SoTA	+3.4	+3.2	-0.8	+3.0	+2.3	+1.4

The **best** and 2nd best results are highlighted.

prediction even without invariant grounding, leaving TIGV with a relatively moderate performance gain.

Second, we find our method performs well on the reason-based question ($Q \rightarrow AR$: Acc@P +8.7, Acc@C +11.8), where the prediction is formulated as a two-step solution that requires both correct answering and a proper reason. This can serve as a test-bed to examine whether a model leans on the spurious correlation, (i.e., if a model tends to leverage the spurious correlation, it could exhibit a good answering ability on $Q \rightarrow A$, but is unable to provide a correct reason on $Q \rightarrow AR$). For TIGV, we notice that it brings much more improvement on $Q \rightarrow AR$ than it does to $Q \rightarrow A$, which means its predictions are well supported with a rational reason. This justifies the proposed invariant grounding is able to rule out the spurious correlation and make a reliable prediction via causal information.

2) *Results on NExT-QA*: As shown in Table III, TIGV demonstrates a remarkable improvement on NExT-QA (Acc@All: +3.0), which validates the effectiveness of our design. Scrutinizing its sub-question type, we notice that the overall improvement originates from the superiority in causal question (Acc@C: +3.4) and temporal question (Acc@D: +3.2). The reasons can be summarized as follows: 1) Causal question generally covers the video content with a long time span, which requires more introspective grounding of the causal scene. Fortunately, IGV’s design philosophy adheres to such demand by eliminating a huge proportion of trivial scenes that is irrelevant to answering

the casual question, thus making the prediction more reliable. 2) Temporal questions tend to include a temporal indicative phase (e.g. “*at the end of the video*”) that serves as a strong signal for grounding indicator to locate the target window.

Apart from that, we are also aware of the slight performance drop on descriptive question (Acc@D: -0.8) when compared with VGT [9]. This is because VGT utilizes additional object feature (i.e., ROI from pre-trained object detector) as the fine-grain attributes of the instances, which benefits its reasoning on descriptive questions. However, the acquisition and utilization of object feature requires complex pre-processing with heavy computation, which inevitably involves more hyper-parameters that demand careful tuning (i.e., number of objects). As a comparison, TIGV only incorporates the clips-level feature to attain efficiency, while also achieving a favorable overall performance.

3) *Results on MSVD-QA and MSRVTT-QA*: To further investigate TIGV’s reasoning ability on descriptive questions, we conduct experiments on MSVD-QA and MSRVTT-QA datasets that focus on the recognition of the visual objects and attributes. The results are shown in the last two columns of Table III. For both datasets, TIGV achieves the leading performance over SoTA. Such improvement is underpinned by the logic that answering descriptive questions requires scrutiny of the causal scene, instead of a holistic view of the entire sequence. It is worth noticing that, although TIGV falls slightly behind SoTA on the descriptive question of NExT-QA (cf. Section V-A2), it prevails on MSVD-QA and MSRVTT-QA datasets. We attribute such discrepancy to the dataset volume. In fact, MSVD-QA and MSRVTT-QA has much more descriptive QA pairs than that of NExT-QA (cf. Table I), which benefits TIGV’s transformer architecture that is known to be data hungry [43].

B. Ablation Study

1) *Contributions of Loss Components*: In the top part of Table IV, we first acquire a baseline model by removing the grounding indicator and scene interverter from TIGV. Then, we exhaust the combination of TIGV’s loss components and design three variants: $\mathcal{L}_{\hat{c}}$, $\mathcal{L}_{\hat{c}} + \mathcal{L}_{\hat{t}}$ and $\mathcal{L}_{\hat{c}} + \mathcal{L}_{v^*}$. Our observations are as follow:

- Using $\mathcal{L}_{\hat{c}}$ solely, which can be viewed as a special case of ERM-guided attention, hardly outperforms the baseline, because grounding indicators can not identify the causal

TABLE IV
ABLATION STUDY ON TIGV

Variants	NExT-QA				MSVD
	Acc@C	Acc@T	Acc@D	Acc@All	
Baseline	53.6	54.0	60.1	54.7	41.6
$\mathcal{L}_{\hat{c}}$	53.5	53.0	61.0	54.6	41.2
$\mathcal{L}_{\hat{c}} + \mathcal{L}_{\hat{i}}$	53.6	54.4	61.4	55.1	42.2
$\mathcal{L}_{\hat{c}} + \mathcal{L}_{v^*}$	54.0	54.7	61.3	55.4	42.1
Baseline+IGV	54.7	55.3	61.1	56.0	42.6
w/o $\mathbf{x}$	54.9	55.1	61.8	56.1	42.6
TIGV	55.0	56.3	62.9	56.7	43.1

scene without clip-level supervision. Such an expected result reflects our motivation on the intervention design.

- $\mathcal{L}_{\hat{c}} + \mathcal{L}_{\hat{i}}$ and $\mathcal{L}_{\hat{c}} + \mathcal{L}_{v^*}$ matched equally in accuracy while consistently surpass baseline and $\mathcal{L}_{\hat{c}}$ in all cases. Such progress shows the effectiveness of the intervention strategy and the introspective regularization imposed on the complement scene.
- The TIGV, which is $\mathcal{L}_{\hat{c}} + \mathcal{L}_{\hat{i}} + \mathcal{L}_{v^*}$, further boosted the above performances significantly. This indicates $\mathcal{L}_{\hat{i}}$ and $\mathcal{L}_{v^*}$ contribute from different aspects and their benefits are mutually reinforcing.

2) *TIGV* versus *IGV*: To demonstrate the benefits of TIGV as integrated instantiation, we follow the general IGV framework (cf: Section IV-C) and create a variant (denoted as “Baseline+IGV” in Table IV) by using TIGV’s baseline as a conventional VideoQA backbone. We notice that the integrated TIGV outperforms the “Baseline+IGV” implementation on both datasets, and the reasons can be provided from two aspects:

- *The Grounding Indicator*: In the preliminary IGV framework, the causal and complement grounding leverage the same global question representations as query vector (cf. (23) and (24)), so that the grounding of different scenes can only rely on their loss components. However, TIGV explicitly models the semantic difference of causal and complement scene by adopting a question embedding and a learnable token (cf. (8)). We experimentally show the advantage of this modification by replacing the TIGV’s learnable token $\mathbf{x}$ (cf. (8)) with $\mathbf{q}_{cls}$ (denoted as “w/o $\mathbf{x}$ ”), where the performance drop indicates the semantic difference of query vectors benefits the TIGV with a better grounding.
- *The Scene Intervener*: Unlike the IGV that performs intervention on raw video clips, TIGV intervenes on the encoded visual token that involves the question semantics. Therefore, it enlarges the sampling space by considering the combination of questions and visual clips, thus, resulting in a more diverse intervention.

C. Backbone Agnostic

In addition, we also validate the generality of the IGV framework by adopting two SoTA models as backbone — the graph-based HGA [26] and memory-based Co-Mem [6]. Apart from that, we also design a simple yet effective architecture on top of HGA as an additional backbone (denoted as HGA*). See Appendix Section C for the detailed architecture, available

TABLE V
APPLY IGV TO SoTA METHODS. “+IGV” DENOTED OUR STRATEGY IS INCORPORATED. ON NExT-QA, ACC@ALL IS REPORTED

Models	MSVD-QA		NExT-QA	
	Baseline	+IGV	Baseline	+IGV
Co-Mem [6]	34.6	37.7	48.5	50.7
HGA [26]	35.4	38.8	50.0	51.7
HGA*	36.1	40.8	50.2	51.3

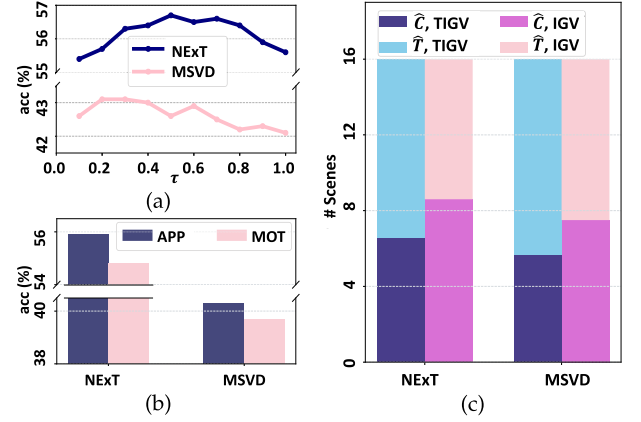


Fig. 7. (a) Study of temperate τ in grounding indicator. (b) Feature analysis. (c) The number of grounded causal and complement scenes.

online. In Table V, we witness a constant improvement for each backbone. Through row-wise inspection, we notice that for each benchmark, IGV brings considerable improvement across different backbone models (+3.1~4.7 for MSVD-QA, +1.1~2.2 for NExT-QA). Such stable enhancement strongly verifies that the IGV framework is orthogonal to backbone design, thus helping to boost the off-the-shelf SoTAs without compromising their underlying architecture.

D. Model Analysis

1) *Study of Grounded Scenes*: To grasp the learning pattern of TIGV, we inspect the number of causal and complement scenes on two datasets, and deliver the statics in Fig. 7(c). As a comparison, we also draw such statics using our preliminary IGV framework (using HGA* as the backbone).

The results on both datasets demonstrate that TIGV is able to pinpoint a small proportion of video tokens as causal scene, while leaving the rest bulk as complement scene. Compare to the preliminary IGV work, TIGV achieves better performance with fewer causal scene, which indicates such integrated instantiation manages an effective invariant grounding by allocating more concentrated answer-critical information. Moreover, we observe that answering on NExT-QA generally takes more causal scene than it does on MSVD-QA. This is because the NExT-QA focuses on relational reasoning among multiple entities, which is typically associated with more visual scenes. Whereas the MSVD-QA emphasizes the description of a single object, where the visual clue can be covered by fewer scenes. Such observation also validates the proposed invariant grounding is operated in an adaptive manner.



Fig. 8. Visualization of prediction on NExT-QA. The green and pink windows indicate the results of the TIGV and preliminary IGV, respectively. The ground-truth answer is highlighted in red.

2) *Study of Hyper-Parameter*: To validate the sensitivity of TIGV against the hyper-parameters, we conduct experiments with variations of τ — the temperature of Gumbel-Softmax in the grounding indicator (cf. (11)). According to Fig. 7(a), we observe that the performance reaches its peaks around 0.5 and 0.2 for NExT-QA and MSVD-QA, respectively. Specifically, compare to MSVD-QA, fluctuation on NExT-QA is more drastic, where tuning on τ causes a 2% difference in their accuracy. This can be attributed to the temporal complexity of NExT-QA, and the oracle length of the causal scene in such a composite scenario certainly requires more careful tuning. It’s noteworthy that, under all tests, TIGV outperforms the baseline by a large margin, which indicates its robustness against variation of hyper-parameters.

3) *Study of Feature*: By convention, we also study the effect of the input condition by ablation on the visual feature. Particularly, we denote *APP* for tests that adopt only the appearance feature as input and *MOT* for tests that utilize the motion feature alone. Fig. 7(b) delivers results on two benchmarks.

We observe that the performance delivered by *APP* is typically better than *MOT*, which implies the invariant grounding draws more on the appearance feature than the motion counterpart. Meanwhile, considering the causal nature of TIGV, we conclude that static correlation tends to bias more in appearance feature. In addition, we also notice that the standalone performances of *APP* and *MOT* can achieve even better results than the full-featured SoTA methods, which demonstrates the robustness of our design.

E. Qualitative Analysis

By nature, TIGV is empowered with visual explainability, and is apt to account for the right scene for its prediction. Following

this essence, we grasp its learning insight by inspecting some predictions together with the grounded causal scene. The results of IGV (using HGA* as the backbone) are also provided for comparison.

In the first row of Fig. 8, we notice that TIGV is able to correctly pinpoint the causal scene via a comprehension of the phrase “for the second time”, then draws out the correct prediction accordingly. However, IGV fails on a reasonable grounding, thus yielding an improper result. In the second row, although both models yield the answer correctly, the grounded scene presented by TIGV is more concise and rational, whereas the IGV covers almost the entire video. We also investigate a failure case in the last row, where the “grass” scene misleads the prediction via a spurious correlation between “grass” and “dog” (i.e., there are many training instances involving “dog with grass”). Since the “grass” scene appears throughout the entire video, our clip-level intervention is unable to rule out the spurious correlation nested within each frame. Worse still, the answer-critical “ball” information is hardly detectable in the clip-level feature, which makes the correct answer “follow the ball” untraceable.

VI. RELATED WORKS

Video Question Answering (VideoQA): Aiming to answer the question in a video scenario, VideoQA is defined as an escalation of imageQA, because the temporal nature of the input has enriched its reasoning process as well as the answer space. Previous efforts towards VideoQA establish their contribution on either a better multi-modal interaction or stronger video representation. Specifically, early studies tend to impose sophisticated cross-modal fusion via attention [4], [26], [44] or dynamic memory [3], [5], [6]. Whereas the recent approaches

turn to model video as a cross-modal graph or a multi-level hierarchy. In the graph-based methods, [26] and [27] first build their typologies based on the heterogeneity of input modality, while [9] enabling relation reasoning in different feature granularity. As for the hierarchical designs, [29] first build a bottom-up pathway by stacking conditional relation block at different level, the following works [7], [8] extend this philosophy by incorporating the object feature. Despite their effectiveness, their visual-explainability still dwells on ERM-guided attention weights, which only reflect the intensity of feature-prediction correlation.

Transformer: Although first introduced as a machine translation technique [45], the transformer has soon underpinned many other communities. As one of its profound practices in computer vision, DETR [22] proposes a *query-based* scheme that enables the object detection to be optimized in end-to-end manner. It adopts a set of learnable object queries as the candidate target, which is fed into a classic transformer decoder for the target generation. Beyond the image field, VisTR [46] brings the framework to video instance segmentation, MDETR [47] makes its extension as a multi-modal object detector. transformer has soon underpinned multi-modal communities, due to its unified structure for both vision and language reasoning. In VideoQA, although some prior works [9], [41] have incorporated a transformer layer, they only use it as a cross-modal fusion block in the encoder, the true power nested in the *query-based* decoder is left untouched.

Invariant Learning: Multi-modal datasets tend to display inherent bias in some forms [48], [49], [50]. In contrast to over-arching reality, the collection process [33], [34] degrades its generalization ability by introducing undesirable correlations between the inputs and the ground truth annotations. To overcome such correlation, invariant learning is developed to discover causal relations from the causal factors to the response variable, which remains constant across distributions. As the most prevailing formulation, IRM [18] promotes this philosophy from feature level to representation level by finding a data representation Υ , from which the optimal predictor φ can yield the prediction $\Upsilon \circ \varphi$ that is stable across all environments. In terms of environment acquisition, previous studies either manually partition the training set by prior knowledge [2], or generates data partition iteratively via adversarial environment inference [19], [51]. Our method, instead of partitioning the training, assumes no prophets about environments but performs causal intervention to perturb the original distribution. To the best of our knowledge, our design is the first to introduces invariant learning framework to VideoQA.

VII. LIMITATIONS

In addition to the technical advantages, we discuss the inadequacy of IGV and TIGV in this section, whose refinement should further boost the performance. 1) Unlike existing SoTA (e.g. HOSTR [7] and VGT [9]) that utilize the object-level feature, TIGV and IGV merely exploit the power of the frame and clip-level features. This partly explains why the performance of IGV and TIGV are limited on the descriptive questions, which

require more detailed object information. Therefore, we expect the incorporation of a fine-grained feature on the object level can mitigate this limitation. 2) Current intervention strategy can possibly threaten causal prediction by introducing context with new shortcuts. Since the scene intervener samples context stratification in a random manner, it is likely to draw undesired substitution that is nested with strong relation towards a wrong answer, thus inducing a false prediction.

VIII. CONCLUSION

In this paper, we pinpoint that the spurious visual-linguistic correlations in VideoQA are triggered by question-irrelevant scenes. In this regard, we first proposed a novel invariant grounding framework IGV, and further integrated it with transformer-empowered instantiation TIGV, which discovers the causal scene and emphasizes its causal effect on the answer. With the grounding indicator and scene intervener, our designs capture the causal patterns that remain stable across complements. Extensive experiments show the superiority of TIGV architecture and the effectiveness of IGV framework. Our future work includes: 1) the spurious correlations can nest in entities, and object-level invariant learning is promising to alleviate this issue; and 2) as the current intervention strategy might threaten the causal prediction by introducing a complement scene with new shortcuts, we will further explore new intervention methods.

REFERENCES

- [1] X. Wang et al., "Reinforced cross-modal matching and self-supervised imitation learning for vision-language navigation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 6629–6638.
- [2] P. Anderson et al., "Vision-and-language navigation: Interpreting visually-grounded navigation instructions in real environments," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 3674–3683.
- [3] C. Fan, X. Zhang, S. Zhang, W. Wang, C. Zhang, and H. Huang, "Heterogeneous memory enhanced multimodal attention model for video question answering," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 1999–2007.
- [4] X. Li et al., "Beyond RNNs: Positional self-attention with co-attention for video question answering," in *Proc. AAAI Conf. Artif. Intell.*, 2019, pp. 8658–8665.
- [5] D. Xu et al., "Video question answering via gradually refined attention over appearance and motion," in *Proc. 25th ACM Int. Conf. Multimedia*, 2017, pp. 1645–1653.
- [6] J. Gao, R. Ge, K. Chen, and R. Nevatia, "Motion-appearance co-memory networks for video question answering," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 6576–6585.
- [7] L. H. Dang, T. M. Le, V. Le, and T. Tran, "Hierarchical object-oriented spatio-temporal reasoning for video question answering," in *Proc. Int. Joint Conf. Artif. Intell.*, 2021, pp. 636–642.
- [8] J. Xiao, A. Yao, Z. Liu, Y. Li, W. Ji, and T.-S. Chua, "Video as conditional graph hierarchy for multi-granular question answering," in *Proc. AAAI Conf. Artif. Intell.*, 2022, pp. 2804–2812.
- [9] J. Xiao, P. Zhou, T.-S. Chua, and S. Yan, "Video graph transformer for video question answering," in *Proc. Eur. Conf. Comput. Vis.*, Springer, 2022, pp. 39–58.
- [10] T. Schuster, D. J. Shah, Y. J. S. Yeo, D. Filizola, E. Santus, and R. Barzilay, "Towards debiasing fact verification models," in *Proc. Conf. Empir. Methods Natural Lang. Process. 9th Int. Joint Conf. Natural Lang. Process.*, 2019, pp. 3417–3423.
- [11] J. Pearl et al., *Models, Reasoning and Inference*, vol. 19, Cambridge, U.K.: Cambridge Univ. Press, 2000.
- [12] J. Pearl, M. Glymour, and N. P. Jewell, *Causal Inference in Statistics: A Primer*. Hoboken, NJ, USA: Wiley, 2016.

- [13] A. S. Ross, M. C. Hughes, and F. Doshi-Velez, "Right for the right reasons: Training differentiable models by constraining their explanations," in *Proc. Int. Joint Conf. Artif. Intell.*, 2017, pp. 2662–2670.
- [14] L. Chen, X. Yan, J. Xiao, H. Zhang, S. Pu, and Y. Zhuang, "Counterfactual samples synthesizing for robust visual question answering," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 10797–10806.
- [15] J. Xiao, X. Shang, X. Yang, S. Tang, and T.-S. Chua, "Visual relation grounding in videos," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 447–464.
- [16] K. Xu et al., "Show, attend and tell: Neural image caption generation with visual attention," in *Proc. 32nd Int. Conf. Mach. Learn.*, 2015, pp. 2048–2057.
- [17] A. Vaswani et al., "Attention is all you need," in *Proc. 31st Int. Conf. Neural Inf. Process. Syst.*, 2017, pp. 5998–6008.
- [18] M. Arjovsky, L. Bottou, I. Gulrajani, and D. Lopez-Paz, "Invariant risk minimization," 2019, *arXiv:1907.02893*.
- [19] T. Wang, C. Zhou, Q. Sun, and H. Zhang, "Causal attention for unbiased visual recognition," 2021, *arXiv:2108.08782*.
- [20] D. Krueger et al., "Out-of-distribution generalization via risk extrapolation (rex)," in *Proc. 38th Int. Conf. Mach. Learn.*, 2021, pp. 5815–5826.
- [21] Y. Li, X. Wang, J. Xiao, W. Ji, and T.-S. Chua, "Invariant grounding for video question answering," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 2928–2937.
- [22] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis.*, A. Vedaldi, H. Bischof, T. Brox, and J. Frahm, Eds., 2020, pp. 213–229.
- [23] Z. Gao, L. Wang, B. Han, and S. Guo, "AdaMixer: A fast-converging query-based object detector," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 5354–5363.
- [24] J. Li, L. Niu, and L. Zhang, "From representation to reasoning: Towards both evidence and commonsense reasoning for video question-answering," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 21241–21250.
- [25] J. Xiao, X. Shang, A. Yao, and T. Chua, "NEXt-QA: Next phase of question-answering to explaining temporal actions," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 9777–9786.
- [26] P. Jiang and Y. Han, "Reasoning with heterogeneous graph alignment for video question answering," in *Proc. AAAI Conf. Artif. Intell.*, 2020, pp. 11109–11116.
- [27] J. Park, J. Lee, and K. Sohn, "Bridge to answer: Structure-aware graph interaction network for video question answering," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 15521–15530.
- [28] D. Huang, P. Chen, R. Zeng, Q. Du, M. Tan, and C. Gan, "Location-aware graph convolutional networks for video question answering," in *Proc. AAAI Conf. Artif. Intell.*, 2020, pp. 11021–11028.
- [29] T. M. Le, V. Le, S. Venkatesh, and T. Tran, "Hierarchical conditional relation networks for video question answering," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 9969–9978.
- [30] J. Jiang, Z. Chen, H. Lin, X. Zhao, and Y. Gao, "Divide and conquer: Question-guided spatio-temporal contextual attention for video question answering," in *Proc. AAAI Conf. Artif. Intell.*, 2020, pp. 11101–11108.
- [31] J. Wang, B. Bao, and C. Xu, "DualVGR: A dual-visual graph reasoning unit for video question answering," 2021, *arXiv:2107.04768*.
- [32] J. Pearl, *Causality: Models, Reasoning and Inference*, 2nd ed., Cambridge, U.K.: Cambridge Univ. Press, 2009.
- [33] A. Torralba and A. A. Efros, "Unbiased look at dataset bias," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2011, pp. 1521–1528.
- [34] W. Chao, H. Hu, and F. Sha, "Being negative but constructively: Lessons learnt from creating better visual question answering datasets," in *Proc. Annu. Conf. North Amer. Chapter Assoc. Comput. Linguistics*, 2018, pp. 431–441.
- [35] P. He, X. Liu, J. Gao, and W. Chen, "DeBERTa: Decoding-enhanced bert with disentangled attention," in *Proc. Int. Conf. Learn. Representations*, 2021.
- [36] E. Jang, S. Gu, and B. Poole, "Categorical reparameterization with gumbel-softmax," in *Proc. Int. Conf. Learn. Representations*, 2017.
- [37] J. Tian, C. Kang, and J. Pearl, "A characterization of interventional distributions in semi-Markovian causal models," in *Proc. AAAI Conf. Artif. Intell.*, 2006, pp. 1239–1244.
- [38] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, pp. 1735–1780, 1997.
- [39] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. Annu. Conf. North Amer. Chapter Assoc. Comput. Linguistics*, 2019, pp. 4171–4186.
- [40] L. Peng, S. Yang, Y. Bin, and G. Wang, "Progressive graph attention network for video question answering," in *Proc. 29th ACM Int. Conf. Multimedia*, 2021, pp. 2871–2879.
- [41] M. Peng, C. Wang, Y. Gao, Y. Shi, and X. Zhou, "Multilevel hierarchical network with multiscale sampling for video question answering," in *Proc. Int. Joint Conf. Artif. Intell.*, 2022, pp. 1276–1282.
- [42] Y. Li, X. Wang, J. Xiao, and T. Chua, "Equivariant and invariant grounding for video question answering," 2022, *arXiv:2207.12783*.
- [43] Y. Liu, E. Sangineto, W. Bi, N. Sebe, B. Lepri, and M. D. Nadai, "Efficient training of visual transformers with small datasets," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, pp. 23818–23830.
- [44] K. Zeng, T. Chen, C. Chuang, Y. Liao, J. C. Niebles, and M. Sun, "Leveraging video descriptions to learn video question answering," in *Proc. 31st AAAI Conf. Artif. Intell.*, 2017, pp. 4334–4340.
- [45] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 5998–6008.
- [46] Y. Wang et al., "End-to-end video instance segmentation with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 8741–8750.
- [47] A. Kamath, M. Singh, Y. LeCun, G. Synnaeve, I. Misra, and N. Carion, "MDETR - Modulated detection for end-to-end multi-modal understanding," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2021, pp. 1760–1770.
- [48] A. Anand, E. Belilovsky, K. Kastner, H. Larochelle, and A. C. Courville, "Blindfold baselines for embodied QA," 2018, *arXiv:1811.05013*.
- [49] J. Thomason, D. Gordon, and Y. Bisk, "Shifting the baseline: Single modality performance on visual navigation & QA," in *Proc. Annu. Conf. North Amer. Chapter Assoc. Comput. Linguistics*, 2019, pp. 1977–1983.
- [50] A. Rohrbach, L. A. Hendricks, K. Burns, T. Darrell, and K. Saenko, "Object hallucination in image captioning," in *Proc. Conf. Empir. Methods Natural Lang. Process.*, 2018, pp. 4035–4045.
- [51] E. Creager, J. Jacobsen, and R. S. Zemel, "Environment inference for invariant learning," in *Proc. 38th Int. Conf. Mach. Learn.*, 2021, pp. 2189–2200.
- [52] H. Ben-Younes, R. Cadène, N. Thome, and M. Cord, "BLOCK: Bilinear superdiagonal fusion for visual question answering and visual relationship detection," in *Proc. AAAI Conf. Artif. Intell.*, 2019, pp. 8102–8109.



Yicong Li received the bachelor's and master's degree from the Huazhong University of Science and Technology and Columbia University, respectively. He is currently working toward the PhD degree with Data Science Institute, National University of Singapore, with a research focus on multi-modal learning. His publications include some top conferences, such as CVPR, *IEEE Transactions on Pattern Analysis and Machine Intelligence*, ACM MM, and AAAI.



Xiang Wang received the PhD degree from the National University of Singapore, in 2019. He is now a professor with the University of Science and Technology of China (USTC). His research interests include recommender systems, graph learning, and explainable deep learning techniques. He has published some academic papers on international conferences, such as NeurIPS, ICLR, KDD, WWW, and SIGIR. He serves as a program committee member for several top conferences, such as KDD, SIGIR, WWW, and IJCAI, and invited reviewer for prestigious journals, such as *IEEE Transactions on Knowledge and Data Engineering*, *ACM Transactions on Information Systems*, and *IEEE Transactions on Neural Networks and Learning Systems*.



Junbin Xiao is currently working toward the final year PhD degree with the Department of Computer Science, National University of Singapore. His PhD topic focuses on VideoQA and he has broad interests in cross-modal video understanding. He has published relevant papers in the top-tier conferences, such as CVPR, ECCV, AAAI, ACM MM, and EMNLP. He serves as reviewers for conference: CVPR'23, ICCV'23, AAAI'23, ECCV'22, and journals: *IEEE Transactions on Multimedia* and *ACM Transactions on Multimedia Computing, Communications, and Applications*. He also serves as program committee members for AAAI'23, ACM MM'19 and MM'20. He organized the 1st video relation understanding challenge in conjunction with MM'19.



Wei Ji received the PhD degree in computer science from Zhejiang University, in 2020. He is a research fellow with the School of Computing, National University of Singapore. He has published several papers in top conferences, such as CVPR, ECCV, SIGIR, AAAI, and journals including *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *IEEE Transactions on Image Processing*, and *IEEE Transactions on Cybernetics*. His current research interests include multi-modal learning, vision and language, and cross-modal retrieval.



Tat-Seng Chua received the PhD degree from the University of Leeds, U.K. He is the KITHCT chair professor with the School of Computing, National University of Singapore (NUS). He is also the distinguished visiting professor of Tsinghua University, the visiting Pao Yue-Kong chair professor of Zhejiang University, and the distinguished visiting professor of Sichuan University. He was the founding dean of the School of Computing from 1998 to 2000. His main research interests include unstructured data analytics, video analytics, conversational search and recommendation, and robust and trustable AI. He is the co-director of NExT, a joint research Center between NUS and Tsinghua University, and Sea-NExT, a joint Lab between Sea Group and NExT. He is the recipient of the 2015 ACM SIGMM Achievements Award, and the winner of the 2022 NUS Research Recognition Award. He is the chair of steering committee of Multimedia Modeling (MMM) conference series, and ACM International Conference on Multimedia Retrieval (ICMR) (2015–2018). He is the general co-chair of ACM Multimedia 2005, ACM SIGIR 2008, ACM Web Science 2015, ACM MM-Asia 2020, and the upcoming ACM conferences on WSDM 2023 and The Web Conferences 2024. He serves in the editorial boards of three international journals. He is the co-founder of two technology startup companies in Singapore.